

OF Fields).

The **tag** qualifier specifies a numeric tag value to be used when encoding a particular value. For brevity, the **tag** keyword SHALL be omitted when specifying a tag qualifier. As a special case, the keyword **anonymous** MAY be used to signal a value that SHALL be encoded without a tag.

Matter TLV supports two forms of tags: **Protocol-Specific Tags** and **Context-Specific Tags**. A protocol-specific tag is a colon-separated tuple containing a **protocol-id** and a **tag-num**. Protocol ids MAY also be specified indirectly, by giving the name of a **PROTOCOL** definition (**protocol-name**) located elsewhere in the schema. An asterisk (*) MAY be used as a shorthand to refer to the id of the **PROTOCOL** definition in which the tag qualifier appears. This protocol is referred to as the *current protocol*.

B.5.8.1. Explicit Tags

A **tag** qualifier that appears on a field within a **STRUCTURE** or **FIELD GROUP**, or on an item within a **LIST**, specifies the exact tag to be used when encoding the associated field/item. Such a tag is called an explicit tag, and MAY be either a **context-specific**, **protocol-specific** or **anonymous** (for **LIST**) tag.

If a field or item lacks a **tag** qualifier, then the encoding will use a **default tag** associated with the underlying field type, if such a tag has been specified.

B.5.8.2. Default Tags

A **tag** qualifier that appears on a type definition, or on an alternate within a **CHOICE OF**, serves as a default tag. A default tag is used to encode a value when an explicit tag has not been given in the schema.

For example, a field within a **STRUCTURE** that refers to a type with a default tag will use that tag if no **tag** qualifier has been specified on the field itself. Similarly, **tag** qualifiers that appear on the alternates of a **CHOICE OF** serve as default tags to be used when no other tag has been specified.

Both **context-specific** and **protocol-specific** tags MAY be used as default tags. 'anonymous' tag SHALL NOT be used as default tag.

Default Tag Qualifier Example

```
vendor-ab-prot8 => PROTOCOL [ id 0x00AB0008 ]
{
  ec-pub-key [0x00AB0008:1] => OCTET STRING,    // default protocol-specific tag using
                                              // a numeric protocol id

  ec-priv-key [vendor-ab-prot8:2] => STRUCTURE // default protocol-specific
                                              // tag using a name
  {
    priv-key [1]      : OCTET STRING,          // explicit context-specific tag

    pub-key [2, optional] : ec-pub-key         // explicit tag overrides default tag on
                                              // ec-pub-key
```